



05.03 Lab

Network Devices

Introduction concepts of proxy server



Google as proxy?

- Google is plan to prevent Chrome users to be tracked across websites by anonymizing IP addresses
- IP Protection, previously referred to as "ip-blindness" or "Gnatcatcher," is a proxy system similar to Apple's Privacy Relay.
- It's designed to run Chrome browser connections through two proxies, one operated by Google and one operated by a third-party (eg, Cloudflare), so that the true public IP address of the user is obscured.



Google IP Protection

- When enabled, the feature will route traffic from selected domains through proxies, making users' IP addresses invisible to those domains.
- Google has a list of domains involved that will evolve over time to better protect users.
- IP Protection will be an opt-in feature, initially, so users will be able to choose whether to activate it or not.

What is a proxy server

- A **proxy server** is a system or router that provides a gateway between users and the internet.
- It helps prevent cyber attackers from entering a private network.
- It is a server, referred to as an “intermediary” because it goes between end-users and the web pages they visit online.

How it works

- Because a proxy server has its own IP address, it acts as a go-between for a computer and the internet.
- Your computer knows this address, and when you send a request on the internet, it is routed to the proxy, which then gets the response from the web server and forwards the data from the page to your computer's browser, like Chrome, Safari, Firefox, or Microsoft Edge

Proxy, Data and Privacy

- A proxy server performs the function of a firewall and filter.
- The end-user or a network administrator can choose a proxy designed to protect data and privacy.
- This examines the data going in and out of your computer or network. It then applies rules to prevent you from having to expose your digital address to the world.
- Only the proxy's IP address is seen by hackers or other bad actors.

Proxy server benefits

- **Enhanced security:** Can act like a firewall between your systems and the internet. Without them, hackers have easy access to your IP address.
- **Private browsing, watching, listening, and shopping:** Use different proxies to help you avoid getting inundated with unwanted ads or the collection of IP-specific data. With a proxy, site browsing is well-protected and near impossible to track.

Proxy server benefits (cont.)

- **Access to location-specific content:** You can designate a proxy server with an address associated with another country. You can, in effect, make it look like you are in that country and gain full access to all the content computers in that country are allowed to interact with.
- **Prevent employees from browsing inappropriate or distracting sites:** You can use it to block access to websites that run contrary to your organization's principles. Also, you can block sites that typically end up distracting employees from important tasks. Some organizations block social media sites like Facebook and others to remove time-wasting temptations.

Types of proxies

- **Forward proxy**

- It sits in front of clients and is used to get data to groups of users within an internal network. When a request is sent, the proxy server examines it to decide whether it should proceed with making a connection.
- A forward proxy is best suited for internal networks that need a single point of entry. It provides IP address security for those in the network and allows for straightforward administrative control.
- However, a forward proxy may limit an organization's ability to cater to the needs of individual end-users.

Types of proxies

- **Transparent proxy**

- A transparent proxy can give users an experience identical to what they would have if they were using their home computer. In that way, it is “transparent.”
- Transparent proxies are well-suited for companies that want to make use of a proxy without making employees aware they are using one.
- It carries the advantage of providing a seamless user experience. On the other hand, transparent proxies are more susceptible to certain security threats, such as SYN-flood denial-of-service attacks.

Types of proxies

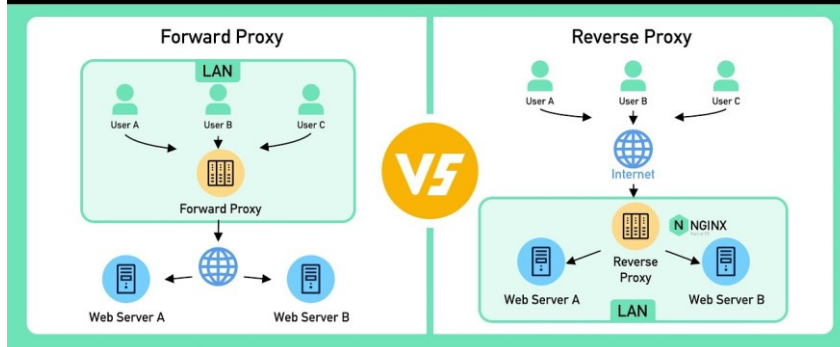
- **Anonymus proxy**
 - An anonymous proxy focuses on making internet activity untraceable. It works by accessing the internet on behalf of the user while hiding their identity and computer information.
 - A anonymous proxy is best suited for users who want to have full anonymity while accessing the internet.
 - While anonymous proxies provide some of the best identity protection possible, they are not without drawbacks. Many view the use of anonymous proxies as underhanded, and users sometimes face pushback or discrimination as a result.

Types of proxies

- **Reverse proxy**
 - Unlike a forward proxy, which sits in front of clients, a reverse proxy is positioned in front of web servers and forwards requests from a browser to the web servers. It works by intercepting requests from the user at the network edge of the web server. It then sends the requests to and receives replies from the origin server.
 - Reverse proxies are a strong option for popular websites that need to balance the load of many incoming requests. They can help an organization reduce bandwidth load because they act like another web server managing incoming requests.
 - The downside in reverse proxies it can potentially expose the HTTP server architecture to an attacker able to penetrate it. This means network administrators may have to beef up or reposition their firewall if they are using a reverse proxy.

Forward vs Reverse Proxy

Forward Proxy vs. Reverse Proxy



SOCKS proxy

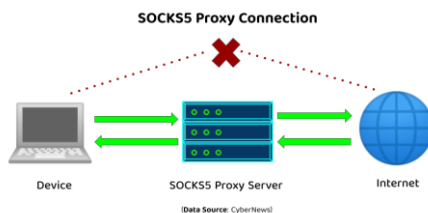
- SOCKS (SOCKS stands for SOCKet Secure) is a proxy which supports User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) connections.
- A SOCKS proxy is not a VPN. The main difference between the two is that in addition to hiding your original IP address, VPN encrypts your data but a SOCKS proxy does not.
- With SOCKS proxy you can:
 - Bypass internet blocks by rerouting your traffic
 - Have reliable connection and performance because of using UDP
 - Faster download speeds
 - Reroute any type of traffic

SSH SOCKS proxy

- SSH solves the lack of encryption in the SOCKS proxy.
- Once you have your SOCKS proxy, you can set up an SSH tunnel between your computer and the proxy server to encrypt and secure your data to keep it safe from many cyber threats.

Set up a SSH SOCKS proxy

- We need:
 - A server with the sshd daemon active (we can download OpenSSH and use it)
 - An Internet connection
 - A client machine with ssh
 - A browser that supports SOCKS (e.g. Mozilla Firefox)



Server and sshd

- Some Linux distros disable tunneling and proxying by default:
 - care must be taken to not expose services and hosts that we might prefer to keep closed to unwanted access.
- The enablement of *sshd*, the daemon that serves ssh sessions, is done by editing the *sshd_config* file.
- Its location varies a little but is usually on */etc/ssh* or */etc/openssh*.

Forward dynamic tunnels

- Any connections using SOCKS protocol to the binding port will be forwarded to the SSH server using its own IP address. To do that, we would use:
 - `ssh -D [bind_address:]port -Nn -f [user@]remote_ssh_server`
 - -D, port-forward the remote server to a local port
 - -N, does not execute any commands on the remote server
 - -f, runs the ssh client in the background

Forward application socket

- Still from the client, we open our favorite browser, for example Mozilla Firefox, and configure the options relating to the proxy server:
 - Let's open the Preferences
 - Let's go to Advanced → Network
 - Click on Settings...
 - We insert the host and port in the fields relating to Host SOCKS (in our case localhost and the port)
 - We select SOCKS v5 and Remote DNS

HTTP(s) proxy

- The HTTP proxy protocol routes client requests from web browsers to the internet
- To perform this operation, the **HTTPS proxy** performs an attack known as Man-in-The-Middle (MITM).
- Some browsers and applications generate alarms stating that the connection is not secure; this is because, for the client, the certificate is generated by the HTTPS proxy itself (i.e. Auth pages of Captive portals).

HTTP(s) vs SOCKS

- They are both types of proxy servers, but they work differently and are used for different purposes.
- HTTP proxies are used to relay HTTP (and HTTPS) traffic. They work at the application layer of the OSI model, specifically the HTTP protocol, HTTP proxies are typically used to bypass censorship or access geo-restricted content.
- SOCKS proxies are a lower-level type of proxy that work at the transport level of the OSI model. SOCKS proxies can be used to handle any type of traffic, not just HTTP. They are typically used to bypass firewalls or access network resources on a restricted network.
- To summarize, HTTP proxies are used for HTTP(s) traffic and SOCKS proxies are used for all types of traffic.

How to configure a HTTP proxy in your browser

- Setting up a HTTP(S) proxy on your browsers is very simple.
- Some can directly manage the settings, others forward to the systems proxy settings.
- You can follow the instructions on this web page to try to surf the web via web proxies
 - <https://www.digitalcitizen.life/how-set-proxy-server-all-major-internet-browsers-windows/>